

YOUTUBE CHANNEL

S KHAN ACADEMY

Date: _____ Math 643 MCQ

① _____ is used to create symbolic

matlab

② ☒ syms ② function ③ symbols ④ syntax

② ☒ sign is used to write equation

① = ☒ == ③ < = ④ > =

③ $dy/dt = ay$; $y(0) = 2$

① $y(2) = 0$ ☒ $y(0) == 2$ ③ $(y_0) == 2$

④ $y(0) = 2$

④ Matlab standard solver for

ordinary difference is the _____

☒ ode45 ② Solve ③ eig ④ none of these

Today Paper

Day. MCOS Math 641310

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i) Command is used to
write differential equation
diff Diff inte

ii) $gn [t, y] = ode45(odefun, tspan, y_0)$
 $y = ?$

function interval Condition

iii) In $[t, y] = ode45(odefun, tspan, y_0)$
 $odefun = ?$

function interval Condition

iv) function $[x, y] = f(x, s)$
 $x = s + x;$

$y = 2 + x + 3^x s;$

end

x is — ?

dependent ~~int~~ input output

V) In $[L, U] = \text{odd}$ **YOUTUBE CHANNEL**
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$t \text{ span} = ?$ ✓

function interval condition

vi) function $[x, y] = f(x, S)$

$$X = S + x;$$

$$Y = 2 + x + 3 * S;$$

end

S is **input**

vii) Syms

$$f = S * \sin(S * x)$$

$$\text{diff}(f, 9)$$

4th

✓
5th

6th

8th

viii) $f = 5 * x^6 + x^4 + 6$

$$d(x) = \text{diff}(f, 5)$$

$$d(2) = 7200$$

ix) In Matlab-Command is used
 for Calculat

✓
 rref reduced echelon form, rxf echelon

Mathlab

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Final terms Quiz

Q: 1

In Mathlab we use for diff

- a) ode
- b) ODE
- ☒ c) diff
- d) Diff

Q: 2

D _____ is

- ☒ a) eigen value
- b) eigen vector
- c) eigen value & eigenvector
- d)

Q: 3

if $f(a) \neq f(c) > 0$ the

- ☒ a) $a = c$
- b) $c = a$
- c) $b = a$
- d)

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Q: 4

function $n[x, y]$ = new function
(a, b) then x is

- ☒ a) out put
- b) in put
- c) out put & in put
- d)

Q: 5

in bisection method

- a) $f(a) * f(b) > 0$
- b) $f(b) * f(a) > 0$
- ☒ c) both
- d)

Q: 6

function $n[x, y]$ = new function
(a, b) then y is

- a) in put
- ☒ b) out put
- c) both.

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S K H A N A C A D E M Y

① In Matlab - we use f for diff

① Code ② ODE ③ ☒ diff ④ Diff

① [P, D] P — and

② D — is

① eigenvalue ② eigen vector

③ ☒ eigen value and eigenvector

③ if $f(a) * f(c) > 0$ the

① ☒ $a=c$ ② $c=a$ ③ $b=c$

④ Function $[x, y] = \text{newfunction}(a, b)$

then x is

① ☒ out put ② Input ③

output and input

⑤ In bisection method

① $f(a) * f(b) > 0$ ② $f(b) * f(c) > 0$

③ ☒ both

⑥ function $[x, y] = \text{newfunction}$
 $[a, b]$ then y is

① ☒ out put ② ☒ out put ③ both

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MCQs

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function $[x, y] = f(x, s)$

$$x = s + y$$

$$y = 2x + 3s$$

and $s =$ _____

a) input b) output c) name of function

syms $u(t) = v(t) =$ _____

ode1 = diff $u(t) =$ _____

a) dsolve(ode) = (usol(t), vsol(t))

b) $s =$ dsolve(ode)

$s \cdot u$

$s \cdot v$

c) both

syms x

$$f = 5 \sin 5x$$

diff(b, 9) will calculate

_____ of function $f = 5 \sin 5x$

a) 9th b) double c) triple

syms x

$$f = 5 * x^6 + x^2 + 6;$$

$$d(x) = \text{diff}(f, 5)$$

$$d(2)$$

a) 5th derivative b) 3rd c) 7200 d) all

1st order initial value problem

a) Euler b) trapezoidal rule

c) Simpson's 1/3 d) Simpson's 3/8

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- To find root of Bisection method in $[a, b]$ we take $c = \frac{a+b}{2}$ if c is positive where $f(a) = -iv$, $f(b) = +iv$

Then which of following is correct

- a) $a = c$ ✓ b) $b = c$ c) $a = b$ d) $c = a$

- For initial value problem $\frac{dy}{dx} = 2xy$, $y(1) = 1$ using step size $h = 0.05$

how many iteration are required

for approximation $y(1.2)$?

- a) 1 b) 2 c) 3 ✓ d) 4

- Solve $\int_0^2 x^2 dx$ by trapezoidal rule

$n = 10$ $h = ?$

- a) 0 b) 0.2 ✓ c) 10 d) 2

- In MATLAB _____ command is use for calculate _____

✓ (rref, rref) echlon)

b) (rref, echlon)

[P, D] 2 mcs

P = Eigen vector ہے P کے

اور ایس میں D کے

D = Eigen value

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- In $[t \ y] = \text{ode45}(\text{odefun}, t_{\text{span}}, y_0)$
 - independent
 - dependent
 - constant
 - non-linear
- MATLAB standard solver for _____ in ode45.
 - linear
 - non-linear
 - Ordinary differential eq
 - Partial differential equation
- Ordinary differential equation implement _____ method variable line step for coefficient computation.
 - Euler
 - R-K method
 - improve Euler
 - Simprun
- Script file save with _____ name as the function name.
 - Same
 - different
 - same or different
 - function
- function $[x \ y] = f(x, s)$

$$x = s + r$$

$$y = 2x + 3s$$

end _____ $y =$ _____

 - input
 - Output
 - name of function

function area = problem

$$\text{area} = x^2$$

end

will not give vale save with name

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